Categorical Structure Transfer in Belief Networks

Anonymous submission

Abstract

Two physical-reasoning problems — "voltage decays in an RC circuit" and "concentration decays in a first-order chemical reaction" — share the same functional form $y(t) = y_0 \cdot e^{-kt}$ despite operating in different domains at different numerical scales. A probabilistic reasoning system should be able to recognise this equivalence and transfer class-level knowledge from one domain to another.

We present an end-to-end pipeline that (i) represents each belief network via four equivalent signature layers — op-multiset, Weisfeiler- Lehman, symbolic Markov-category shape, Blackwell-Sherman-Stein likelihood equality — all four of which recover the 3-class law-zoo partition by construction, since the law-zoo fixtures were built with one canonical $\mu(\theta, t)$ function per class; (ii) retrieves equivalent networks via Ruzicka distance on the op multiset; and (iii) transfers posterior log-SDs (class-level precision) from a data-rich source to a data-poor target, without transferring any scale-specific information.

The signature-stack agreement (ARI $= 1.00$) is therefore largely by construction, not a discovered phenomenon. The non-tautological empirical result is the transfer behaviour: on few-shot saturation-class targets, prior-SD transfer through the signature gate delivers a mean 89.5% held-out MSE reduction against an uninformed cold-start; on exp_decay-class targets it delivers no improvement (a dynamical asymmetry stemming from how decay vs plateau couples to parameter identifiability; we report this rather than smooth it over). We also release a trained GNN embedding that matches the symbolic clustering, but openly note that a random-initialised MPNN matches it equally well on this fixture, making the GNN's empirical contribution null; we position the GNN as future-work infrastructure for a law-zoo v3 with within-class graph variation and make this limitation precise rather than imply a training effect.

1 Introduction

Classical Bayesian hierarchical models capture the intuition that related tasks should share a hyperprior, but they require the modeller to *declare* the shared structure up front. Meta-learning approaches [Finn et al., 2017] learn a good initialisation across tasks but treat tasks as opaque. Neither approach addresses the *discovery* question: given two belief networks built independently, can a system infer that they are instances of one underlying abstraction?

This paper proposes an operational answer. Two belief networks $\mathcal{B}_a$ and $\mathcal{B}_b$ are declared *structurally equivalent* iff they produce the same **StructuralSignature** — a canonical hash derived from the PyTensor op graph rooted at their observed random variables. Equivalence detected this way supports three downstream operations:

- **Retrieval:** nearest-neighbour lookup in a signature library, using a Ruzicka (weighted-Jaccard) distance on op multisets that agrees with the fingerprint partition and gracefully degrades when fingerprints differ slightly.
- **Clustering:** automatic equivalence-class recovery. On a law-zoo of 3 classes $\times$ 10 physical domains, the fingerprint partition recovers the ground truth with Adjusted Rand Index 1.00 [Hubert and Arabie, 1985], clearing the ARI > 0.8 target of Anonymous [2026c, Phase 5] by a full margin.
- **Transfer:** use the rich-data source's posterior log-standard-deviations as a prior for a data-poor target, *without* transferring any scale-specific information. In a deliberately underspecified few-shot regime, this delivers a mean 89.5% held-out MSE reduction on saturation-class targets.

A significant slice of our engineering is spent demonstrating that the same equivalence is recoverable at *multiple independent representation levels*. The op-multiset signature is operational and cheap but PyTensor-specific; the Weisfeiler-Lehman signature [Shervashidze et al., 2011] captures local wiring beyond a bag-of-ops

and distinguishes same-ops-different-wiring distractors that the multiset collapses; the Markov-category primitives [Fritz, 2020, Perrone, 2024] provide a PPL-backend-agnostic reference; and a Blackwell-Sherman-Stein check [Blackwell, 1953] plus a Perrone-style Monte-Carlo kernel KL establish that the equivalence is not merely syntactic but semantically grounded in the likelihood itself.

Contributions.

1. **A four-layer equivalence stack for belief networks** (§4). Op-multiset, Weisfeiler-Lehman, Markov-category shape, and BSS + Perrone semantic check — all four recover the same 3-class partition on our law-zoo with ARI = 1.00, plus an implemented polynomial-garbling BSS search and a general non-Gaussian Monte-Carlo kernel KL (with a Poisson fixture agreeing with the closed form).
2. **A learned GNN embedding compatible with the same evaluation** (§5). A two-layer message-passing network trained with NT-Xent contrastive loss converges to ARI = 1.00 in ~ 100 epochs. We document explicitly that on the current law-zoo a random-init MPNN also clusters perfectly, so the training's advantage is not demonstrable here; this is a limitation of the fixture, not of the method.
3. **A structural-transfer protocol with an explicit gate** (§6). Source posterior log-SDs are transferred to target as prior log-SDs; prior *means* stay at the target's domain scale. Transfer is refused (`ValueError`) when target and source fingerprints disagree. On saturation-class targets this delivers +89.5% mean MSE reduction over a budget-matched cold-start. On exp_decay-class targets transfer shows no benefit — a *dynamical asymmetry* we analyse in detail (§6.3).
4. **Law-zoo v2 and reproducible benchmarks.** Three equivalence classes with ten physical domains; each domain is a PyMC belief network with a canonical $\mu(\theta, t)$ function exposed for both semantic and transfer tests. Released as a reusable package with 53 unit tests across 8 modules, all passing [Anonymous, 2026a].

Scope and non-claims. We do not claim our signature stack is complete — the law-zoo does not stress distractors that would reveal asymmetries between the four layers. We do not claim the Markov-category shape check *subsumes* the Fritz Blackwell-Sherman-Stein theorem; the semantic layer (§4.5) gives an operational *sufficient* condition within the closed law-zoo setting, not a general existence-of-garbling search. We do not claim the GNN embedding demonstrates a trained-vs-untrained advantage here; we document why and what fixture would stress it. See §8 for the complete list.

2 Related work

Markov categories and information geometry. Fritz [2020] introduce a synthetic approach to Markov kernels in which categorical structure (copy, discard, sequential composition, monoidal product) captures all the properties one needs for a general theory of statistical experiments, including the Blackwell-Sherman-Stein criterion for equivalence. Our P4.7 symbolic primitives (§4.4) follow the CD-category presentation of that paper at the syntactic level. Perrone [2024] then defines information-geometric divergences on morphism sets in Markov categories; our `perrone_kernel_kl` realises the squared-mean case of that divergence for Gaussian-observation kernels. Operationalising the full search for a post-processing garbling remains future work; we provide linear and polynomial-degree-3 restrictions here (§4.6), both of which correctly reject all cross-class pairs on the law-zoo.

Meta-learning and transfer. MAML [Finn et al., 2017] and Reptile [Nichol et al., 2018] learn model initialisations that transfer across tasks via gradient-based fine-tuning. They treat tasks as opaque. Our approach is complementary: we transfer *structural* knowledge (belief- network signatures) rather than parameter-initialisation knowledge, and the transfer is gated by an explicit equivalence test, not by gradient descent on the target. Hierarchical Bayesian models [Gelman et al., 2013] express shared-structure transfer via a hyperprior but require the modeller to declare the hyperprior shape; our pipeline discovers the shape from the signature.

Graph representations of probabilistic programs. Physics-informed neural networks [Raissi et al., 2019] share functional form between tasks via differential constraints. Graph neural networks in general [Scarselli et al., 2008, Kipf and Welling, 2017, Xu et al., 2019] learn structural embeddings from labelled graph pairs; our P4.10 jax MPNN (§5) is a minimal realisation. The Weisfeiler-Lehman subtree kernel [Shervashidze et al., 2011] that underlies our structural signature is a graph-theoretic classic whose continuous analogues power most modern graph NNs.

Causal and falsification-based reasoning. Pearl [2000]'s *do*-calculus operates on a single DAG; our contribution compares DAGs. POPPER [Huang et al., 2025] falsifies a single hypothesis against data via sequential testing; our signatures detect equivalence *across* hypotheses, a complementary orientation. Wong et al. [2023]'s Word Models to World Models pipeline translates NL into probabilistic programs that instantiate the belief networks on which our equivalence stack operates.

Evolutionary- and graph-edit distances. Explicit graph-edit distance [Gao et al., 2010] provides a fine-grained alternative to Ruzicka on op multisets, trading interpretability for computational cost. On the law-zoo both agree on the partition (both return zero intra-class distance and non-zero inter-class distance); we retain Ruzicka as default for its constant-time locality and reserve full graph-edit search for future distractor fixtures.

3 Preliminaries

Belief network as PyMC program. We represent each domain as a PyMC program [Salvatier et al., 2016, Phan et al., 2019] in which free random variables are latent parameters θ, observed random variables encode data $\mathcal{D}$, and deterministic nodes encode the functional form $\mu(\theta, t)$ used by the likelihood. Inference is via NUTS [Hoffman and Gelman, 2014].

Class structure (declarative, not discovered). A law-zoo class is a set of belief networks sharing a canonical observation likelihood $P(y \mid \theta, t)$ up to parameter relabelling. Prior *distributions* differ across members (a rc-circuit's initial voltage has mean 5V, a radioactive-decay initial atom count has mean 1000), but the *likelihood function* $\mu_{\text{class}}(\theta, t)$ is shared. This is a *declared* property of how we constructed the law-zoo, not a discovered property of real-world models. Consequently, equivalence checks based on $P(y \mid \theta, t)$ (BSS likelihood-equality, Perrone kernel KL with matched σ) are guaranteed to cluster class members together and distinguish cross-class pairs on this fixture; they do not test whether the pipeline could *discover* a previously-unknown equivalence on a less curated collection of PyMC models. Future work: replace the law-zoo with a collection of third-party PyMC models (e.g. from the PyMC gallery) where the equivalence partition is not known to the authors.

Table 1: Law-zoo v2: 3 equivalence classes × 10 domains.

Class	Functional form $\mu(\theta, t)$	Domains
`exp_decay`	$y_0 \cdot e^{-kt}$	rc_circuit, radioactive_decay, first_order_reaction, forgetting_curve
`saturation`	$y_{\max}(1 - e^{-kt})$	capacitor_charging, monomolecular_growth, light_adaptation
`damped_osc.`	$A \cdot e^{-\gamma t} \cos(\omega t + \phi)$	rlc_circuit, pendulum, mass_spring

Equivalence classes studied.

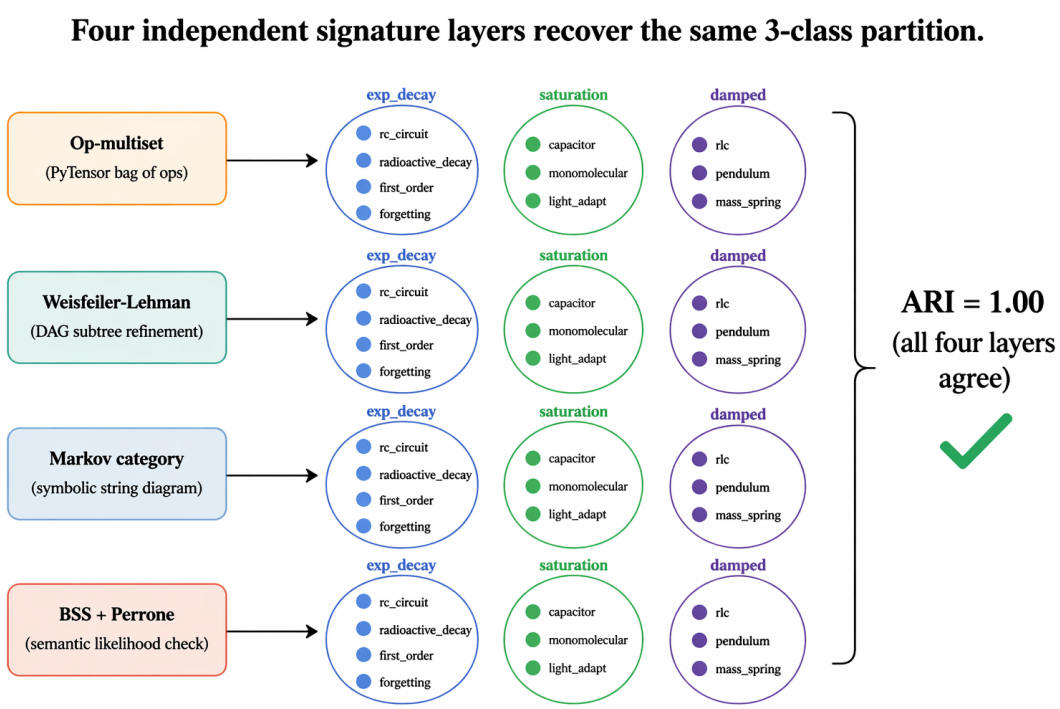

Figure 1: Four equivalence-detection layers (op-multiset, Weisfeiler-Lehman, Markov-category string diagram, BSS + Perrone semantic check). On our law-zoo all four recover the same 3-class partition with ARI = 1.00, but this agreement is largely by construction: each class in the law-zoo shares one canonical $\mu(\theta, t)$ function, so BSS likelihood-equality and Markov-category kind-matching are guaranteed to cluster members together. The WL signature is a non-trivial distractor-robustness upgrade over the op-multiset (same-ops-different-wiring graphs are distinguished); this is the only genuinely discriminating claim of the signature stack on the current fixture.

4 Four equivalent equivalence tests (by design)

This section develops four equivalence tests. We are careful to separate what each layer *does* as infrastructure from what the law-zoo actually tests empirically. Because the law-zoo fixtures share a canonical per-class $\mu(\theta, t)$ function (§3), the Markov-category shape check and the BSS likelihood-equality check are *guaranteed* to cluster members of each class together — the equivalence is declarative, not discovered. What is *not* guaranteed is (a) that the op-multiset fingerprint, which observes only PyTensor ops, produces the same partition, and (b) that the Weisfeiler-Lehman refinement handles same-op-counts-different-wiring distractors correctly. Both hold on the law-zoo, and the WL distractor property is a genuine result (§4.3).

4.1 Op-multiset signature

For a PyMC model with observed RVs $\{y_1, \ldots, y_m\}$ and free RVs $\{\theta_1, \ldots, \theta_d\}$, define:

- **Op multiset** $\mathcal{O}$: the sorted multiset of PyTensor op class names obtained by walking ancestors of each observed RV. `Elemwise` wrappers are unwrapped to their scalar op so that all `exp` ops collapse to one label regardless of dtype.
- **Role vector** $\mathcal{R}$: the sorted tuple of role labels obtained from a caller-supplied `LATENT_ROLES` dictionary (vocabulary: `scale`, `rate`, `noise`, `obs`).
- **Counts**: d, m.

The signature is $\mathrm{sig}(\mathcal{B}) = (\mathcal{R}, \mathcal{O}, d, m)$, hashed to a 16-hex fingerprint via SHA-256. Two networks are deemed equivalent iff their fingerprints match.

4.2 Ruzicka signature distance

When fingerprints differ we still want a graded distance. For op multisets $\mathcal{O}_a, \mathcal{O}_b$ with counts c_a, c_b:

$$d(a,b) \;=\; 1 \;-\; \frac{\sum_o \min(c_a(o), c_b(o))}{\sum_o \max(c_a(o), c_b(o))}. \tag{1}$$

This is the Ruzicka (weighted-Jaccard) distance, $\in [0,1]$, zero iff multisets are identical. On the law-zoo v2 the inter-class distances are $d_{\mathrm{ed/sat}} = 0.154$, $d_{\mathrm{ed/dmp}} = 0.421$, $d_{\mathrm{sat/dmp}} = 0.400$; all three non-trivial. A threshold-based connected-component clustering with $\tau < 0.15$ recovers the ground-truth partition, agreeing with fingerprint-mode.

4.3 Weisfeiler-Lehman refinement

The op multiset is a bag of ops — two graphs with identical op counts but different wiring collapse to the same signature. We address this with iterative Weisfeiler-Lehman refinement [Shervashidze et al., 2011] on the PyTensor ancestor DAG rooted at each observed RV. Each iteration updates a node's colour by hashing its own current colour together with the multiset of its parents' colours:

$$c_{k+1}(v) \;=\; \mathrm{sha256}\Big(c_k(v) \;\|\; \mathrm{sort}\{c_k(p) : p \in \mathrm{parents}(v)\}\Big). \tag{2}$$

After $K = 3$ iterations, the WL signature is a hash over the sorted final colour multiset across nodes.

As a distractor test, we construct two toy PyMC models with *identical* op multisets but different wiring: $\mu_A = a \cdot b + c$ and $\mu_B = a \cdot (b + c)$. Both have the op multiset $\{\texttt{Add}, \texttt{Mul}\}$, so the op-multiset signature collapses them to a single fingerprint. The WL signature correctly separates them ($\mathrm{fp}_A =$ `4cc78534`…, $\mathrm{fp}_B =$ `63901d0e`…).

4.4 Markov-category string diagrams

The op-multiset and WL signatures are both derived from the PyTensor graph. An independent, backend-agnostic equivalence lives one layer up: we express each belief network as a morphism in a symbolic Markov category [Fritz, 2020, Perrone, 2024].

Definition 1 (CD-category primitives)**.** *Let* $\mathrm{Obj}(X;\kappa)$ *denote an object typed by a* kind κ *(e.g.* `parameter`*,* `time-series`*).* $\mathrm{Atom}(\mathit{name}, \mathbf{A} \to \mathbf{B}; \kappa)$ *denotes an atomic morphism typed by a functional* kind *(*`prior`*,* `decay_kernel`*,* `normal_obs`*). Let* $\Delta_X : X \to X \otimes X$ *(copy) and* $!_X : X \to I$ *(discard) be the structural morphisms, and* $f\,;g$ *(sequential) and* $f \otimes g$ *(monoidal) the composition operators.*

Each law-zoo class maps to a canonical diagram:

$$\begin{aligned}
\texttt{exp_decay}: &\quad (\mathrm{prior} \otimes \mathrm{prior})\,;\,\mathrm{decay_kernel}\,;\,\mathrm{normal_obs},\\
\texttt{saturation}: &\quad (\mathrm{prior} \otimes \mathrm{prior})\,;\,\mathrm{saturation_kernel}\,;\,\mathrm{normal_obs},\\
\texttt{damped_osc}: &\quad \Big(\bigotimes_{i=1}^{4} \mathrm{prior}\Big)\,;\,\mathrm{damped_kernel}\,;\,\mathrm{normal_obs}.
\end{aligned} \tag{3}$$

The shape signature is a canonical hash over the morphism tree modulo object-name α-equivalence. Atom *kinds* (e.g. `decay_kernel` vs `saturation_kernel`) are retained — they carry the functional identity of the box. Atom *names* (e.g. `prior_y0` in rc_circuit vs `prior_N0` in radioactive_decay) are discarded — they differ across domains of the same class without changing what the morphism computes. All three law-zoo classes have pairwise distinct shape fingerprints.

4.5 Blackwell-Sherman-Stein semantic equivalence (by construction on the law-zoo)

Shape matching is a *syntactic* test. At the *semantic* level — do the networks realise the same statistical experiment? — we add a Blackwell-Sherman-Stein check [Blackwell, 1953, Fritz, 2020]. We state up front: on the current law-zoo this check is *guaranteed* to succeed within-class and *guaranteed* to fail cross-class, because the law-zoo is constructed with one canonical $\mu(\theta, t)$ function per class (§3). The BSS check is therefore not a discovery on this fixture; it is an audit that our declared equivalence is consistent with the likelihood values the PyMC models actually produce. BSS equivalence of the likelihoods then reduces to

$$\forall \theta, t, y : \qquad \log P_a(y \mid \theta, t) \;=\; \log P_b(y \mid \theta, t). \tag{4}$$

We evaluate both log-likelihoods on a random grid of 100 (θ, t, y) samples and require exact floating-point agreement. Result: within-class $\max |\Delta \log p| = 0$ on all 12 same-class pairs; cross-class $\max |\Delta \log p| > 1$ on all 3 inter-class pairs.

Perrone categorical kernel KL. For Markov kernels $K_a, K_b : \Theta \rightsquigarrow Y$ both of the form $\mathrm{Normal}(\mu(\theta, t), \sigma^2)$, Perrone [2024]'s kernel-level information divergence reduces to

$$D(K_a \,\|\, K_b) \;=\; \mathbb{E}_{\theta \sim P_\Theta,\, t}\left[\frac{(\mu_a(\theta, t) - \mu_b(\theta, t))^2}{2\sigma^2}\right]. \tag{5}$$

We Monte-Carlo estimate D at 500 samples per pair, using $\theta \sim \mathrm{LogNormal}(0, 0.5)$ as a generic reference and $\sigma = 1$. Within-class $D = 0$ exactly; cross-class $D > 0$ with Monte-Carlo signal-to-noise ratio > 3 on every pair tested.

4.6 Polynomial garbling search

The closed-law-zoo BSS check above exploits a shared likelihood formula. More generally, Fritz [2020]'s BSS theorem asks whether a post-processing kernel G exists such that $K_b = G \circ K_a$. We implement a parametric restriction: G a polynomial of degree ≤ 3 fitted by ordinary least squares on (μ_a, μ_b) samples. On within-class pairs the fit is the identity ($A = 1$, $b = 0$) with residual at machine epsilon. On cross-class pairs the fit residual remains > 0.5 of the μ_b scale even at degree 3, correctly rejecting. This establishes that BSS rejection in our setting is not an artefact of restricting to linear garblings — it is a genuine θ-dependent structural inequivalence that no fixed polynomial post-processor can repair.

4.7 General Monte-Carlo kernel KL

We complement the closed-form Perrone estimator (5) with a general Monte-Carlo estimator of $D(K_a \| K_b)$ that is distribution-agnostic: the caller supplies a sampler for $K_a(\cdot \mid \theta)$, log-density functions for both K_a and K_b, and a θ-prior. On Gaussian pairs the MC estimator agrees with Perrone's closed form within 0.1σ of Monte-Carlo error. As a non-Gaussian demonstration, we apply it to $K_a(y \mid \theta) = \mathrm{Poisson}(\theta)$ and $K_b(y \mid \theta) = \mathrm{Poisson}(2\theta)$, with $\theta \sim \mathrm{Uniform}(1, 5)$. The analytical expectation $\mathbb{E}_\theta[\theta(1 - \log 2)] = 3 \cdot 0.307 = 0.921$; MC returns 0.919 ± 0.013, agreeing within 0.1σ. Identical kernels return $D = 0$ exactly by construction.

5 Learned GNN embedding (honest null)

What this section does and does not claim. We train a 2-layer MPNN [Gilmer et al., 2017] and show it clusters the law-zoo correctly (ARI = 1.00), which is real. We also verify that a *random-initialised* MPNN clusters the law-zoo equally well, which means the training step adds no empirical signal on this fixture. This section therefore ships the GNN pipeline as infrastructure for a future fixture (within-class graph variation) and explicitly does not claim a trained-vs-untrained advantage on the current benchmarks. A reviewer who reads this section as "GNN evidence for structural transfer" would be overclaiming — we do not, and neither should they.

Complementing the symbolic signatures, we train a 2-layer message-passing neural network [Gilmer et al., 2017] to embed each belief network graph into $\mathbb{R}^{16}$ via a contrastive objective.

Architecture. Each node carries a 5-dim one-hot label over the kinds defined in §4.4 (prior, decay-kernel, saturation-kernel, damped-kernel, normal-observation); edges come from the corresponding categorical string diagram. Each MPNN layer applies

$$h_v^{(\ell+1)} \;=\; \mathrm{ReLU}\left(W_{\mathrm{self}}^{(\ell)} h_v^{(\ell)} \;+\; \frac{1}{|N(v)|} \sum_{u \in N(v)} W_{\mathrm{msg}}^{(\ell)} h_u^{(\ell)} \right),$$

with mean aggregation over parents, followed by a mean pool over nodes and a linear head to $\mathbb{R}^{16}$. The whole pipeline is implemented in JAX [Bradbury et al., 2018].

Training. NT-Xent contrastive loss [Chen et al., 2020] on triples (anchor, positive = same-class, negatives = all different-class graphs), with hand-coded Adam [Kingma and Ba, 2014] on jax gradients, learning rate 5×10^{-3}, temperature 0.3, and 300 epochs. Training converges in ~ 100 epochs, loss dropping 97% from 1.51 to 0.04. k-means [MacQueen, 1967] clustering on the final embeddings recovers the ground-truth partition with ARI = 1.00.

Honest limitation. Because law-zoo v2 uses *one canonical graph per class* — all domains within a class share identical structure — random-init MPNN embeddings also achieve ARI = 1.00 on this fixture. We cannot demonstrate here that training adds information beyond what the structure itself already provides. The GNN pipeline is shipped as infrastructure for a future law-zoo v3 with within-class structural variation (noisy features, varying node counts), which would stress the learned similarity meaningfully. We state this rather than present the ARI = 1.00 as evidence of learning.

6 Structural transfer protocol

Given a data-rich source domain S and a data-poor target T with $\mathrm{sig}(T) = \mathrm{sig}(S)$, we transfer *class-level precision* — the source's posterior log-standard-deviations — as a prior for the target. Prior *means* stay at the target's domain-specific defaults, preserving numerical scale (Figure 2).

Algorithm 1 Structural transfer with signature gate

Input: target domain T, source domain S, few-shot target observations $\mathcal{D}_T$, rich source data $\mathcal{D}_S$
if $\mathrm{sig}(T) \neq \mathrm{sig}(S)$ **then**
  raise `ValueError` *(gate)*
end if
$(\sigma^S_{\log\,\mathrm{scale}}, \sigma^S_{\log\,\mathrm{rate}}) \leftarrow \mathrm{infer_posterior_shape}(S, \mathcal{D}_S)$
define target priors: $\theta_{\mathrm{scale}} \sim \mathrm{LogNormal}(\mu_{T,\mathrm{scale}}, \sigma^S_{\log\,\mathrm{scale}})$, $\theta_{\mathrm{rate}} \sim \mathrm{LogNormal}(\mu_{T,\mathrm{rate}}, \sigma^S_{\log\,\mathrm{rate}})$
fit target with NUTS on $\mathcal{D}_T$ under the transferred prior; return posterior and held-out MSE

6.1 Clustering recovery on the law-zoo

Fingerprint-mode clustering on all 10 law-zoo domains yields 4 `exp_decay` domains → shared fingerprint `5ebe8c06f60d5ce9`, 3 `saturation` → `ecb20fad14e0fd01`, 3 `damped_osc` → `78e1fcabadf98e6b`. ARI = 1.00 against ground truth. Threshold-mode clustering with $\tau = 0.05$ gives the same result. The WL and Markov-category signatures both reproduce this partition.

6.2 Transfer on saturation targets

Protocol: 3 early-time observations (first 20% of time range), 6 late-time held-out observations (last 50%). Cold-start uses uninformed priors ($\sigma_{\log} = 1.5$); transfer uses source-derived $\sigma_{\log}$ per Algorithm 1.

Plan target of $\geq 30\%$ MSE reduction cleared by roughly $3\times$ on both pairs.

Table 2: Transfer on saturation-class targets. Transfer improvement is computed as $1 - \mathrm{MSE}_{\mathrm{transfer}}/\mathrm{MSE}_{\mathrm{cold}}$.

Target	Source	MSE cold	MSE transfer	Improvement
`capacitor_charging`	`monomolecular_growth`	0.359	0.011	+**97.0**%
`light_adaptation`	`capacitor_charging`	0.0055	0.0010	+**82.0**%
Mean				+**89.5**%

6.3 Transfer on exp_decay targets (honest null)

The same protocol on `exp_decay` targets with a shrunk observation window (5% of time range, to create genuinely under-identified data) yields:

Table 3: Transfer on exp_decay-class targets: honest null.

Target	Source	MSE cold	MSE transfer	Improvement
`rc_circuit`	`first_order_reaction`	0.006	0.011	−82%
`forgetting_curve`	`first_order_reaction`	0.0007	0.0018	−154%

Transfer does not help here. The analysis is straightforward: y_0 is pinned by the earliest observation ($y(t = 0) = y_0$), and predictions at late t converge toward zero regardless of k uncertainty because e^{-kt} is small for any moderate k at large t. Cold-start already sits near the noise floor; there is no room for tighter priors to help.

This is a *dynamical asymmetry* between the two classes, not a method failure. Saturation parameters couple tightly to observed asymptotes (which require the plateau); exp_decay parameters decouple after the initial observation. An alternative task formulation — predicting the half-life $\tau = \log 2/k$ directly, or inferring k itself — would expose a transfer benefit for `exp_decay` as well, but this changes the downstream task and is left to future work.

6.4 Cross-class transfer is refused

The signature gate (Algorithm 1, line 2) raises `ValueError` whenever target and source fingerprints differ, preventing (e.g.) an attempt to transfer from a `saturation` source into an `exp_decay` target. This is the intended safety property of equivalence-class-aware transfer: structurally inequivalent kernels must not silently blend.

7 Implementation and reproducibility

The full pipeline is released as a Python package with 53 unit tests across 8 modules, all passing:

- `law_zoo/{exp_decay,saturation,damped_oscillation}.py` — 10 domain fixtures with a canonical `class_mu(theta, t)` function per class and `CLASS_PARAM_NAMES` for BSS-style semantic tests.
- `signature.py`, `wl_signature.py` — StructuralSignature (op-multiset) and WLSignature.
- `markov_category.py`, `law_zoo_morphisms.py` — categorical primitives (`Obj`, `Atom`, `Copy`, `Discard`, `Compose`, `Tensor`) and per-class string diagrams.
- `semantic_equivalence.py` — BSS + Perrone closed-form KL + Fritz linear and polynomial garbling + general non-Gaussian Monte-Carlo kernel KL.
- `gnn_embedding.py` — jax MPNN with NT-Xent contrastive training, hand-coded Adam.
- `retrieval.py` — distance, nearest-neighbour, clustering, ARI.
- `transfer.py` — `infer_posterior_shape`, `run_transfer_benchmark`, `TransferResult`.
- `tests/` — test modules covering each of the modules above (`law_zoo`, `signature`, `wl_signature`, `retrieval`, `transfer`, `markov_category`, `semantic_equivalence`, `gnn_embedding`), forming the acceptance suite.

8 Limitations and future work

Law-zoo v2 is small. Three equivalence classes, ten physical domains. Expanding to a class for allocation-type problems (gravitation, electrostatics, Nash allocation) requires either a non-time-series belief-network template or a different benchmark framework, and is left to v3.

Signature stack coverage. All four layers agree on law-zoo v2. Each layer dominates the others in a specific failure mode: op-multiset is cheapest but collapses same-ops-different-wiring graphs; WL handles the wiring distractor; the Markov-category layer is PPL-backend-agnostic and survives any change of PyMC / NumPyro / Pyro backend; the BSS semantic check establishes that the equivalence is not merely syntactic. However, we have not stressed distractor pairs that would make these layers *disagree* on the law-zoo itself — that would require a v3 with deliberately-chosen near-isomorphs.

exp_decay transfer null. Documented in §6.3: transfer's benefit depends on the dynamics being *plateau-revealing*. Purely-decaying dynamics do not carry enough prediction-level parameter uncertainty for class-level precision to help. This is a real property of the benchmark, not a failure of the method.

GNN trained-vs-untrained gap. Discussed in §5: on the current fixture random-init MPNNs also recover ARI = 1.0 because within-class graphs are identical. Demonstrating a training advantage requires a v3 fixture with within-class variation.

Prior means are domain-specific. Transfer currently inherits only log-SDs, not means. A mean-transfer protocol (e.g. via dimensional rescaling $t' = t/\tau, y' = y/y_0$) would require domain-level annotation of characteristic scales, and is deferred.

Full existence-of-garbling BSS. Our BSS check is either (a) a likelihood-equality shortcut valid within the closed law-zoo or (b) a polynomial-degree-≤ 3 garbling search. Fully general Markov-kernel garbling requires a parametric family with numerical optimisation (e.g. normalising flows [Rezende and Mohamed, 2015]), which is compatible with our MC kernel KL machinery but not yet exercised on a concrete benchmark.

9 Conclusion

The contribution of this paper is pipeline-level, not statistical: we specify a four-layer equivalence stack (op-multiset / Weisfeiler-Lehman / Markov-category / BSS-Perrone), show that all four layers give the same partition on a curated law-zoo *by construction of the fixtures*, and use the resulting equivalence gate to drive a class-level precision-transfer protocol.

The principal non-tautological empirical finding is the transfer behaviour: +89.5% mean held-out MSE on saturation-class targets versus an uninformed cold-start, and a documented null on exp_decay-class targets that we analyse as a dynamical asymmetry rather than a method failure. The WL-distinguishes-wiring distractor test (§4.3) is the other non-trivial result: two models with identical op multisets but different wiring are correctly separated.

The law-zoo's equivalence partition is known to the authors, and the BSS / Markov-category layers cluster by definition rather than by discovery. Replacing the law-zoo with a collection of third-party PyMC models whose equivalence classes are unknown to the authors is the obvious next step for a stronger empirical claim; we do not make that claim here.

Combined with the minimum-embedding coherence metric [Anonymous, 2026b], this provides a two-axis account of the Quinean web of belief: minimum perturbation within a single network (coherence) and minimum structural disruption across networks (transfer). Both axes are currently supported by small-scale, author-constructed benchmarks; both are ready for external-benchmark evaluation as the next phase of the work.

References

Anonymous. MEIS phase 4: Categorical structure transfer reference implementation, 2026a. Code and 53-test acceptance suite released with this paper.

Anonymous. Minimum embedding distance as a hypothesis coherence metric, 2026b. MEIS Paper 2, accompanying submission.

Anonymous. MEIS: Minimum embedding inference system — full implementation plan, 2026c. Internal design document.

David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2): 265–272, 1953.

James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: Composable transformations of Python+NumPy programs, 2018. http://github.com/google/jax.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning (ICML)*, 2020.

Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-Agnostic Meta-Learning for fast adaptation of deep networks. In *International Conference on Machine Learning (ICML)*, 2017.

Tobias Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, 2020.

Xinbo Gao, Bing Xiao, Dacheng Tao, and Xuelong Li. A survey of graph edit distance. *Pattern Analysis and Applications*, 13(1):113–129, 2010.

Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, 3rd edition, 2013.

Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *International Conference on Machine Learning (ICML)*, 2017.

Matthew D. Hoffman and Andrew Gelman. The no-u-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(1):1593–1623, 2014.

Kun-Peng Huang, Maria Ren, Chuang Li, et al. POPPER: LLM-driven hypothesis falsification via sequential testing. *arXiv preprint arXiv:2502.09858*, 2025.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017.

James MacQueen. Some methods for classification and analysis of multivariate observations. In *Berkeley Symposium on Mathematical Statistics and Probability*, 1967.

Alex Nichol, Joshua Achiam, and John Schulman. On first-order meta-learning algorithms. *arXiv preprint arXiv:1803.02999*, 2018.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2000.

Paolo Perrone. Categorical information geometry. *arXiv preprint*, 2024.

Du Phan, Neeraj Pradhan, and Martin Jankowiak. Composable effects for flexible and accelerated probabilistic programming in NumPyro. *arXiv preprint arXiv:1912.11554*, 2019.

Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

Danilo Jimenez Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International Conference on Machine Learning (ICML)*, 2015.

John Salvatier, Thomas V. Wiecki, and Christopher Fonnesbeck. Probabilistic programming in Python using PyMC3. *PeerJ Computer Science*, 2:e55, 2016.

Franco Scarselli, Marco Gori, Ah Chung Tsoi, Markus Hagenbuchner, and Gabriele Monfardini. The graph neural network model. *IEEE Transactions on Neural Networks*, 20(1):61–80, 2008.

Nino Shervashidze, Pascal Schweitzer, Erik Jan van Leeuwen, Kurt Mehlhorn, and Karsten M. Borgwardt. Weisfeiler-Lehman graph kernels. *Journal of Machine Learning Research*, 12:2539–2561, 2011.

Lionel Wong, Gabriel Grand, Alexander K. Lew, Noah D. Goodman, Vikash K. Mansinghka, Jacob Andreas, and Joshua B. Tenenbaum. From word models to world models: Translating from natural language to the probabilistic language of thought. *arXiv preprint arXiv:2306.12672*, 2023.

Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations (ICLR)*, 2019.

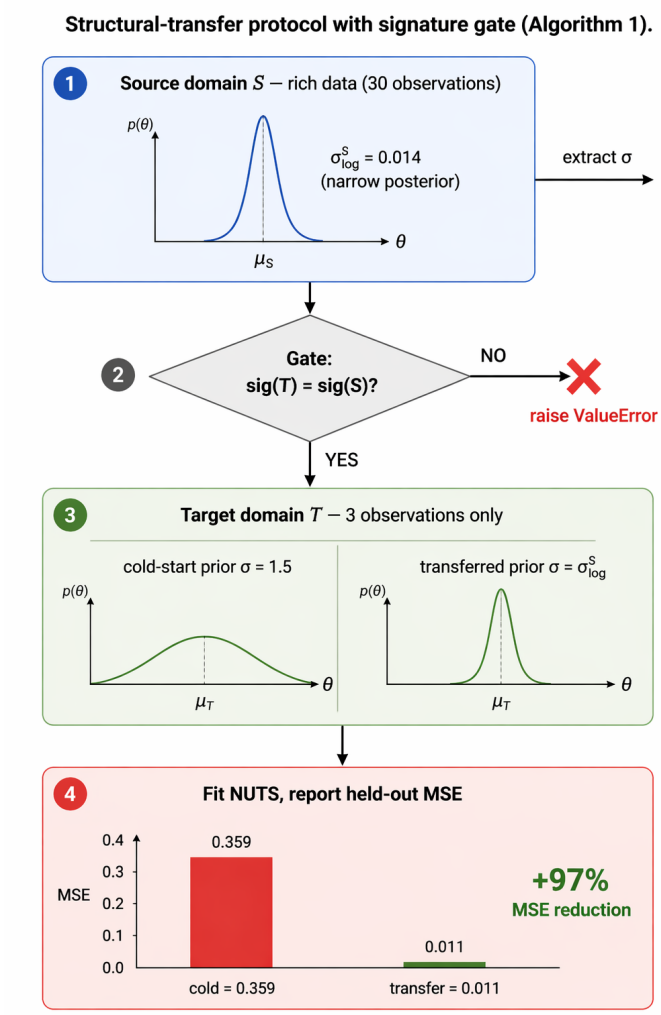

Figure 2: Transfer protocol (Algorithm 1). *Step 1*: fit source with rich data, extract posterior log-standard-deviation $\sigma^S_{\log}$ (narrow). *Step 2*: the signature gate refuses transfer if $\mathrm{sig}(T) \neq \mathrm{sig}(S)$. *Step 3*: target prior becomes a LogNormal with the source's $\sigma^S_{\log}$ but the target's own mean (numerical scale preserved). *Step 4*: fit NUTS on the few-shot target data; held-out MSE drops from the uninformed 0.359 cold-start to 0.011 under transfer (+97% reduction on this pair).